



MERU UNIVERSITY OF SCIENCE AND TECHNOLOGY

BACHELOR OF SCIENCE IN COMPUTER TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE

FILE ENCRYPTION TOOL USING FERNET ALGORITHM

CT206/104329/20: ALVIN OKIYA ANDAYE

A research project Submitted in Partial Fulfillment of the Requirements
of the Bachelor of Science in Computer Security and Forensics of Meru
University of Science and Technology

September, 2023

DECLARATION

I hereby declare that the work presented in this document is my original work with citations and references from different sources and has not been presented and submitted for a degree in any other university. No part of this report shall therefore be duplicated without my prior consent.

Student Reg. No.

Student Name

Signature

Date

CT206/104329/20

ALVIN OKIYA ANDAYE

ABSTRACT

Information security is increasingly becoming a basic requirement for people who use computers and computer devices. However, conflict comes between security vs convenience. Here the tool to be provided will strike a balance between the two providing confidentiality in data and efficiency in its implementation.

Table of Contents

DECLARATION.....	ii
ABSTRACT.....	iii
CHAPTER ONE	1
1.1. Background of study	1
1.2 Motivation for study.....	2
1.3 Problem statement.....	2
1.4 Research objectives	3
1.4.1 General objectives.....	3
1.4.2 Specific objectives	3
1.5 Significance of the study	3
1.6 Scope of the study	4
1.7 Assumptions in the study	4
1.8 Limitations of study	4
CHAPTER TWO: LITERATURE REVIEW.....	5
2.1 Overview	5
2.2 Functions of existing file encryption tools.....	5
2.2.1 VeraCrypt Disk Encryption Tool.....	5
2.2.2 Open SSL	5
2.2.3 An Encryption and Decryption tool safeguarding Patients Health Records	6
2.2.4 GnuPGP (Gnu privacy guard).....	6
2.3 Components the of Existing Encryption and Decryption Tools.....	6
2.4 Features/Characteristics of the existing Encryption and Decryption tools	7
2.6 Related Studies.....	8
CHAPTER 3: METHODOLOGY	9
3.1 Overview	9
3.2 Research Design.....	9
3.3 Population and Sampling	9
3.4 Data collection and instrumentation	10
3.5 Development tools and materials	10

3.6 System Development and Methodology	10
3.7 Data processing and Analysis	11
3.8 Ethical Considerations	12
CHAPTER 4	13
SYSTEM ANALYSIS	13
4.1 FEASIBILITY STUDY	13
4.2 OVERALL DESCRIPTION OF THE CURRENT SYSTEM	13
4.2.1 Judiciary system Use-case Diagram	14
4.2.2 Judiciary system Class diagram	15
4.2.3 Judiciary system sequence Diagram	15
4.2.4 Judiciary System Data Flow Diagram	16
4.2.5 Judiciary System Activity Diagram	17
4.3 Requirement gathering	18
4.3.1 Functional Requirements	18
4.3.2 Nonfunctional requirements	21
4.4 Summary	22
Appendix	23
Questionnaire	23
Feasibility study report for file encryption tool in the Judiciary system set up	24
The purpose of this feasibility report is to address Purpose	24
First solution	24
Second solution	24
Recommendation	25
Research Schedule	25
Budget	26
REFERENCES	26

CHAPTER ONE

1.1. Background of study

In the rapid evolving error of technology, information is becoming one of the most crucial things that we need to secure. Confidentiality being an important aspect of data security is crucial, since data leakage may not only endanger the data owner's privacy but also ruin the reputation of the business as well as violate regulations such as ISO standards for confidentiality ISO 27001:2022 Annex 6:6 (TENDER, 2022). Confidentiality should be guaranteed both during the lifetime of the data (when the data is still in computing devices) and also after it has been deleted, that is it should be irrecoverable after deletion (Zhang et al, 2018). However, this has not been the case as it is common to find deleted data in recycle bins, where they can be accessed by unauthorized users. It is also common to find sensitive data stored in plain text which makes it vulnerable to leakage. Due to this, attackers use this loophole to access data during industrial espionage with the aim of malicious activity to the systems of the targeted organization or to get a competitive advantage based on the acquired information.

The need for confidentiality can be traced back to 500BC how cryptography emerged being only used by militaries with simple encryption methods such as substitution cipher, Spartan encryption roman encryption and World War 2 encryption. Data encryption became popular in the 1970s with the need for information retention by governments (Tresorit Team, 2023). Encryption has evolved to new and more secure methods being used to safeguard confidentiality, maintain authenticity as well as ensure nonrepudiation of data.

The aim of this research is to create an Encryption and Decryption tool that will provide confidentiality (to data being transmitted, stored data and deleted data), Authentication, Integrity of data as well as data Nonrepudiation.

1.2 Motivation for study

The motivation for this study is to offer more resistance to emerging vulnerabilities, from the very basic things such as safe file storage and unrecoverable deletion to safeguard confidentiality. The motivation hence comes since the research will involve designing user-friendly tools that are easy to use and accessible to a broad range of users including those with limited technical experience or expertise. This will thereby strike a balance between security and user-friendliness.

1.3 Problem statement

Due to advancement in technology and cryptography being part of it, the security of encryption is no longer dependent on the encryption method but the secrecy of the keys used (Tresorit TEAM, 2023). In case an attacker is able to access or master the key, no matter how strong the algorithm is, the information is not safe and it can either be stolen or compromised. The larger problem is not only in the using of keys as plaintext, but also the absolute lack of encryption tools in most computers in organizational computers especially those with sensitive information that must be safeguarded. This makes it far much easier for information leakage incase an attacker accesses the computer either physically or remotely, using this as the base of a larger attack.

1.4 Research objectives

1.4.1 General objectives

The objective of this project is to ensure that Meru Law Courts' staff members are able to secure confidential data belonging to both the organization and the clients on both computers that carry sensitive data and shared personal computers.

1.4.2 Specific objectives

- To create a tool that will ensure sensitive or private data stored cannot be readable to any unauthorized persons.
- To create a tool that is compatible with different platforms of operating systems.
- To create a tool that uses an algorithm which ensures secure deletion of data.
- To create an algorithm tool that securely encrypts and decrypts files.
- To test and evaluate the effectiveness of the tool created.

1.5 Significance of the study

The tool is very crucial as it will enable the organization to secure data that is sensitive through storage in encrypted form, security of data in transit and secure deletion of confidential and sensitive organization data that is not supposed to be leaked. The clients coming to file cases to the court will be confident in the security of their files stored online since they can't be compromised by unauthorized persons. This study will ensure that the organization is in accordance with ISO standard for confidentiality, ISO 27001:2022 Annex 6:6 (TENDER, 2022). This will ensure the organization has an added measure to protect confidential information from unauthorized disclosure.

1.6 Scope of the study

This Encryption and Decryption tool will focus mostly on data in personal computers used by staff, researching both on the data that is still stored and is on its lifetime as well as deleted data. The study will involve checking on methods in which are used to store sensitive data, privacy of personal and sensitive data as well as confidentiality of both data in use or deleted data.

1.7 Assumptions in the study

In conducting this study, the following assumptions were made. It was assumed that:

1. Meru university of science and technology prioritizes data security and confidentiality, hence making it necessary to develop a tool that can protect sensitive files.
2. The staff within the organization will comply using the encryption and decryption tool as part of the organization's security policies.
3. The encryption and decryption tool will have user-friendly interface to enable ease of use and adopted easily by staff members.

1.8 Limitations of study

The purposive sampling procedure used decreases the generalizability of findings. Due to this, the study will not be generalizable to all departments of the organization.

CHAPTER TWO: LITERATURE REVIEW

2.1 Overview

Securing information and maintaining data confidentiality are of importance, in today's advancing technology. As technology continues to evolve the demand for reliable encryption tools becomes increasingly crucial. This literature review intends to offer a summary of the methods, algorithms and tools used for encryption and decryption. It will shed light on their strengths, weaknesses and areas that require exploration. The ultimate objective is to provide insights for research endeavors, in developing an innovative and all-encompassing file encryption and decryption solution.

2.2 Functions of existing file encryption tools

In this section we are going to look at some of the projects that have been previously done and their functions. They are as follows

2.2.1 VeraCrypt Disk Encryption Tool

This tool was developed to provide high level security to a wide range of data, being able to encrypt an entire disk using the AES algorithm. However, its complex interface made it intimidating to users with little or no technical experience (Roach, 2023). This makes it difficult to use. It would also slow down the system, hence affecting normal operations.

2.2.2 Open SSL

According to (Ivanovs, 2023), OpenSSL is among the powerful encryption tools that exist, it is widely used by developers, system administrators and security professionals. It has a library that supports a wide range of cryptographic functions such as different encryption algorithms like AES,

DES, RC4 and RSA. Other key features include being able to manage digital certificates and presence of hashing functions such as MD5, SHA-1, SHA-2 and SHA-3, not leaving out SSL/TLS support.

2.2.3 An Encryption and Decryption tool safeguarding Patients Health Records

This is a tool that was created to bridge the gap in confidentiality and privacy of patients' information (Orishaba, 2022). It was to be pegged in as an extension on an already existing EHR (Electronic Health Record) system. This gives patient privacy rights on their information. This privacy rights prevents patients information from being used in inappropriate contexts.

2.2.4 GnuPGP (Gnu privacy guard)

It is a free and open source software tool used to secure data storage as well as communications. According to (Project, 2020), it uses strong encryption algorithms such as AES, Twofish, CAST5 and IDEA. It also uses digital signatures and comes with a key management system that allows users to manage their public and private keys easily. It also has a cross-platform compatibility and is easy to use. The big challenge in it is a steep learning curve for beginners.

2.3 Components the of Existing Encryption and Decryption Tools

User Interface- most have a user interface that is easy to use, however majority are not user-friendly.

Cross-platform compatibility- this allows it to be used with different operating systems.

Encryption algorithm- the existing system have strong encryption algorithms such as AES, RSA to securely encrypt files.

Decryption algorithm- This is usually based on the encryption algorithm.

2.4 Features/Characteristics of the existing Encryption and Decryption tools

According to Ivanovs (2023), there are several features to different existing tools for encryption and decryption. These include:

Strong security features- they use strong encryption algorithm some using symmetric encryption methods such as AES while some use asymmetric encryption, which involves use of both public and private keys.

Cross platform compatibility – a feature allowing them to run on a wide range of operating systems be it Mac Os, Windows, iOS and Linux.

To those using asymmetric encryption, digital certificates are used for authentication purposes.

Two Factor Authentication for added security.

Random Password generation used to protect the file from decryption by unauthorized user.

2.5 Challenges

The current encryption and decryption tools do show and have certain challenges which include:

No Random Initialization Vector (IV) that is used to prevent patterns in encryption operation. Lack of the vector can cause hackers to understand the trends in the patterns giving them an idea of how to study the algorithm and crack it.

Majority of the current encryption tools are not user-friendly, hence making them hard to implement in organizations with people with little or no technical experience.

Striking a balance between robust security and user-friendliness has been a challenge to many of the tools developed. This is the major reason why many secure tools are not user-friendly.

2.6 Related Studies

According to (Orishaba, 2022), her aim of creating the encryption and decryption tool was to combat threats to confidentiality and privacy of patients' data. She went further to categorize these threats into two: organizational threats and systemic threats. Curbing such would therefore make health organizations compliant to privacy and security rules under HIPAA (Luxton, et al., 2012).

Privacy and Confidentiality are therefore even more crucial when it comes to patients' data.

When it comes to forensics, encryption can become a challenge. This is because if the evidence is encrypted, the investigator cannot know what is in it. However linear scanning of the memory can be used to generate trial keys from all possible memory position to decrypt the container (*Recovery of Encryption Keys From Memory Using a Linear Scan*, 2008). This is essential for forensic examiners in recovering encrypted evidence so as to get evidence that can be admissible in a court of law.

CHAPTER 3: METHODOLOGY

3.1 Overview

The proposed File Encryption and Decryption tool is intended to be an application that will be securely encrypting and decrypting files to ensure that confidentiality of sensitive data is not compromised. Due to the increasing use of technology, a large percentage of information belonging to corporates and institution are stored in the computers, therefore if an attacker gains unauthorized access either physically or remotely, there has to be another level of security to deter him from gaining access to confidential and sensitive data that may lead to financial loss or loss of reputation to the company. That is where this tool comes in.

3.2 Research Design

Research design is the overall strategy or simply a structured and systematic approach that one uses as a plan to answer the research question and meet objectives (DeBose, 2018). It constitutes the blueprint for collection, measurement and analysis of data.

In this research of the proposed file encryption and decryption tool, diagnostic research was used to identify the solution of the problem affecting file encryption as well as security of data at rest on computers and then try to come up with a solution of the problem (Bless, 2023). The purpose of this tool is therefore to address the problem and give a solution to the issue. The data collected will therefore assist to tell if the system is supposed to be implemented and how the system will work provided it will fit into the Judiciary security infrastructure.

3.3 Population and Sampling

A population is a complete set group of individuals, and that group comprises a nation or a people having common characteristics (Momoh, 2023). It is important for one to understand this as it relates with sampling.

Sampling is a selection of data from a larger group of data or selection of data from a population (Tuovila, 2020). A sample is usually a representative of the population and hence it has similar properties to the population. Stratified sampling performs better in large scale network of data. In this study, 30 people shall be sampled from the staff team. The law court has different departments, so at least 2 people shall be picked from each department to ensure that all members of the bank are well represented. We will do the sample across the premises giving a total of 16 persons as respondents.

3.4 Data collection and instrumentation

Data collection is the process of gathering data for use in business decision making, strategic planning, research and other purposes (Stedman, 2022). However, according to Ewen (2018), data collection is the process of gathering and measuring information on variables of interest, in an established systematic fashion enabling one in answering stated research questions, evaluate outcomes and test hypothesis.

For this system, questionnaires were used to collect data. Questionnaires is the most appropriate data collection tool for this research because it can be used in areas with large group of respondents and accessing the questionnaires is easy since they can be sent electronically or posted to the respondents and back to the researchers in convenient times giving the respondents the opportunity to freely express their views. The questionnaires are close ended questions and some are open ended questions.

3.5 Development tools and materials

Since this tool is designed to have a cross-compatibility feature, a programming language to be selected should be one that can enable it have the expected features. Due to this, python language is the selected language to be used due to its flexibility, cross-compatibility and ease simplicity. This will enable the tool meet one of its objective that is being compatible with other operating system hence flexible to use in both windows, linux and mac operating systems.

The python IDLE is the coding interface to be used for coding the tool. Since it is easy to use and consumes a small memory in the personal computer.

Hardware requirements include: RAM of at least 4gb laptop with high processing power and monitor resolution.

Software requirements include: latest python IDLE interface, Browser like chrome or brave, windows Operating System and a virtual box with linux operating system to test the cross-compatibility feature.

3.6 System Development and Methodology

According to Hitachi (2023), System development methodology is a methodology for systematically organizing the best Ways to develop systems efficiently, including descriptions of tasks to be performed at each stage of the development process and drafted documents. There are several types of software development methodologies that professionals use in developing, planning, producing and testing systems (Indeed Editorial Team, 2023). System Development Life Cycle (SDLC) is a structured process that enables the production of high-quality, low-cost software, in the shortest possible production time (Synopsis, 2023). It consists of five stages Planning, Analysis, Design, Implementation and Maintenance.

In this research spiral methodology will be applied. This is because it is suitable for large-scale implementation, efficient cost estimation and involves customer feedback. The biggest advantage

of this methodology is its flexibility in that additional functionality can be added at a later date (Kumarurpadhyay, 2020). This made it the most convenient methodology to use as it allows adjustments to be made based on user's feedback.

Due to this questionnaires and stratified sampling are used to get enough information to lead us in gaining information of the current state of data privacy and confidentiality of sensitive data.

Questionnaires are to be used to ensure that information is obtained from employees including that which they can't be confident in saying during interviews due to fear of their identities being disclosed.

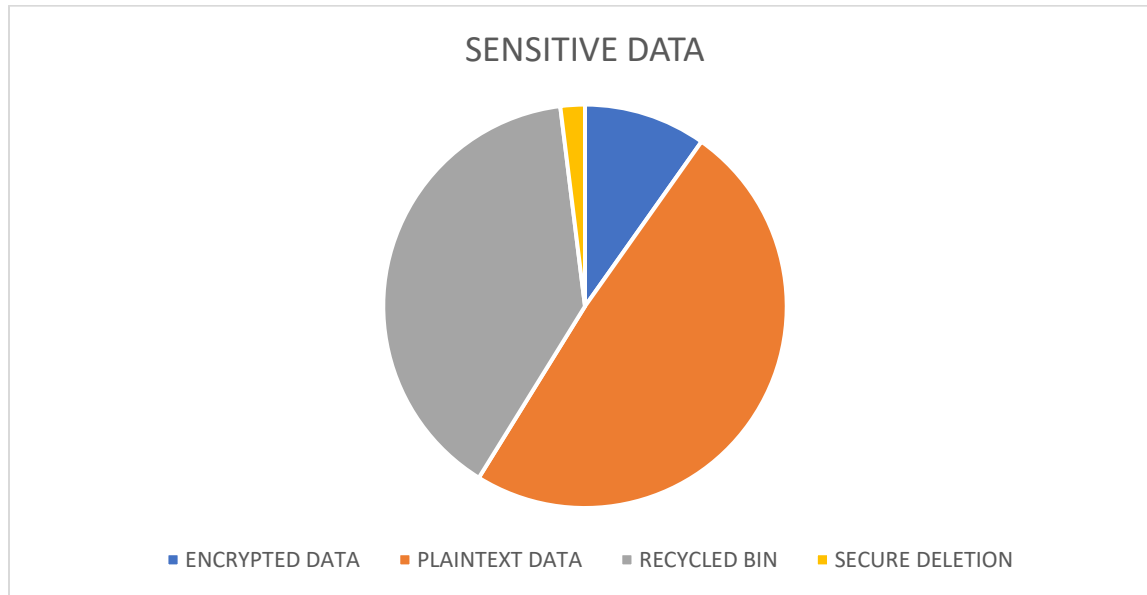
Stratified sampling is to be used as a technique to ensure that all the departments in the institution being studied currently are well represented so as to ensure that as many weaknesses as possible are captured from all the different departments.

3.7 Data processing and Analysis

Data processing is a series of actions or steps performed on data to verify, organize, transform, integrate, and extract data in an appropriate output form for subsequent use (Duggal, 2020). Methods of processing must be rigorously documented to ensure the utility and integrity of the data.

In this case questionnaires were administered, majority of which were closed ended which was to ensure we reach out even to the busiest people so as to ensure the data to be processed is as accurate as it can be. Open ended questionnaires were also used for the Court Administrators and ICT support team to ensure we get detailed feedback from them since they are key stake-holders in computers and the data involved.

After the collection, the data was cleaned and visualized in form of a pie chart to get the difference in the sensitive data such as passwords, personal documents, sensitive organizational and departmental documents which were stored in plaintext versus the encrypted ones, not forgetting data that is assumed to be deleted yet is in the recycle bin, and if accessed by an attacker can be used for malicious purposes. The following was the resultant pie chart.



3.8 Ethical Considerations

Ethical consideration refers to a set of principles that guide your research designs and practices. Among these principles are voluntary participation, informed consent, anonymity, results, communication, potential for harm and confidentiality (Bhandari, 2021).

Statement

I am committed to the highest ethical standards especially in my research on File Encryption and Decryption tool using fernet algorithm.

I am committed to:

- Integrity-I will conduct myself with integrity in my dealings within and outside school.
- Excellence-I will conscientiously strive for excellence in my work

I will be accountable as an individual for my ethical conduct and for compliance with applicable laws

- Respect-I will respect the rights and dignity of others.

Permission

I have a permission of grant from the school through the lecturer in-charge to carry out this research.

CHAPTER 4

SYSTEM ANALYSIS

4.1 FEASIBILITY STUDY

A short assessment of the current methods of storing and deletion of files was done to determine whether developing a new tool is the best solution. An analysis on the resources to be used and the need for them was carried out. In order to accomplish these a feasibility study was carried out. Feasibility study is the assessment of the viability of a project to see whether the project is likely to succeed (Kenton, 2023). The study was aimed at measuring how beneficial the new tool will be to the organization. During the study it was discovered that the organization has enough computers which have enough technical capability to run the proposed tool. Legal feasibility refers to examining all legal requirements and barriers (*Legal Feasibility of Projects*, n.d.). The proposed tool meets all legal and contractual laws of a country. For more details find a feasibility report on appendix 1.

4.2 OVERALL DESCRIPTION OF THE CURRENT SYSTEM

The current system at Meru Law court for filing cases called the Case Tracking System (CTS) is online. A case is first filed physically and a stamp from both the Attorney General and the Magistrate is put on the physical document. From here, the document is scanned in order to create a softcopy which is then sent to the court assistant in-charge who downloads it to his computer then uploads it as an online copy to the Case Tracking System. The physical file is then put on the line up of other filed cases waiting to be given a mention date by the magistrate. From this step henceforth, the outcomes of the case are usually uploaded to the CTS and a copy send to both he accused and the complainant, informing them of the dates and time for the next proceedings or rulings. The system is usually important since in case the original file is missing, the online copy can be printed, and the case continues from the previous steps instead of filing it all over again.

However at the stage of the Court Assistant receiving the document, in case of internet issues, the documents are usually temporarily stored in the Court Assistants' desktop until the internet issue is resolved. This is the stage where the proposed tool is important, since attackers look for such loopholes to compromise integrity and confidentiality of cases, to either make them disappear or tamper with them.

4.2.1 Judiciary system Use-case Diagram

A use-case diagram is a tool mapping the interaction between a system and its users to show the interaction between them (Indeed Editorial Team, 2022). It specifies the event in a system and how those events flow, however, it does not describe how those events are implemented (IBM, 2021). The use-case diagram below will illustrate the handling of documents from the process of filing until the time the defendant is summoned to appear in court.

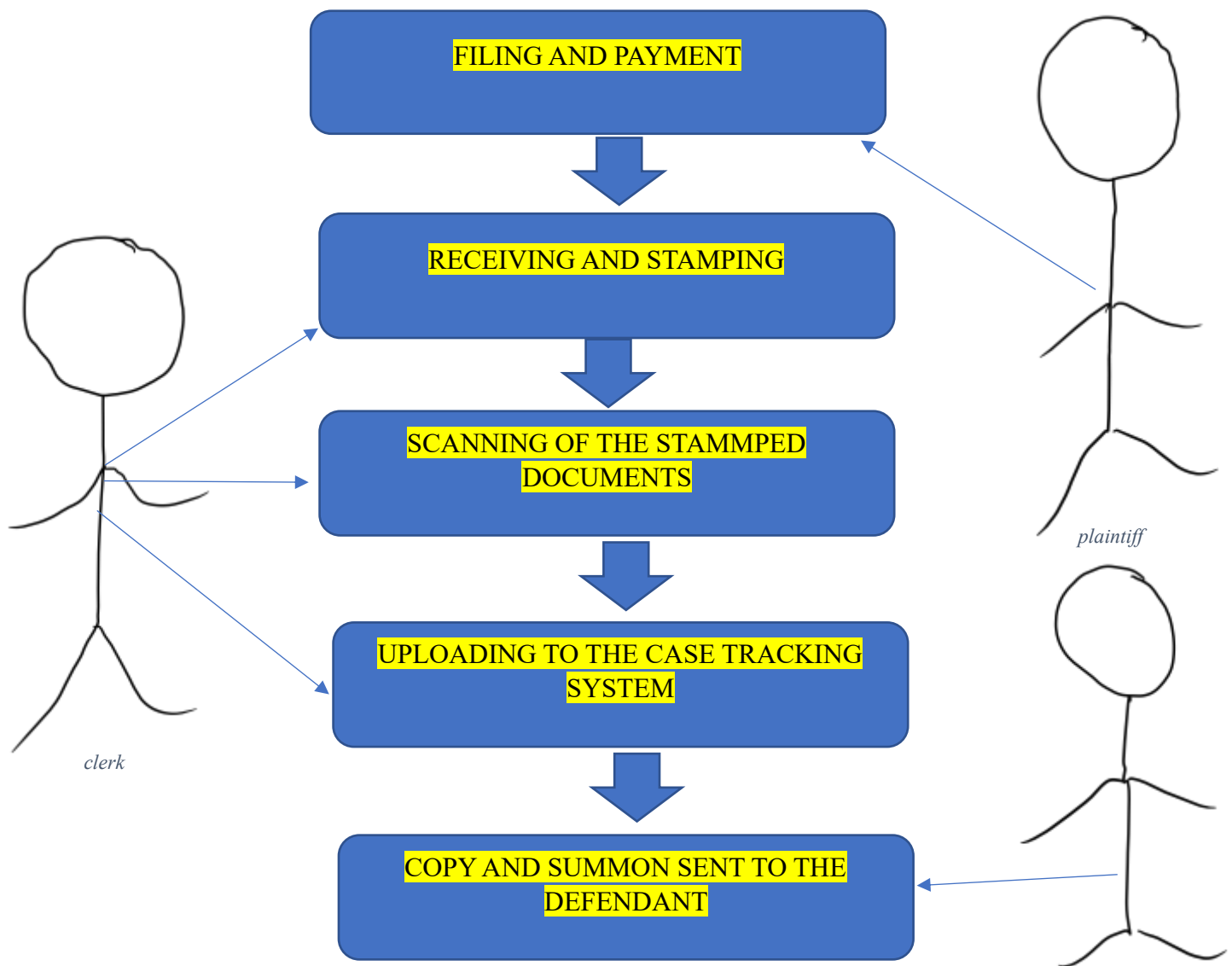


Figure 4.2.1: The current judicial system use-case diagram

4.2.2 Judiciary system Class diagram

Class diagrams are the blueprints of your system. They are used to model the objects that make up the system, to display the relationships between these objects, to describe what those objects do and the services they provide (IBM, 2021).

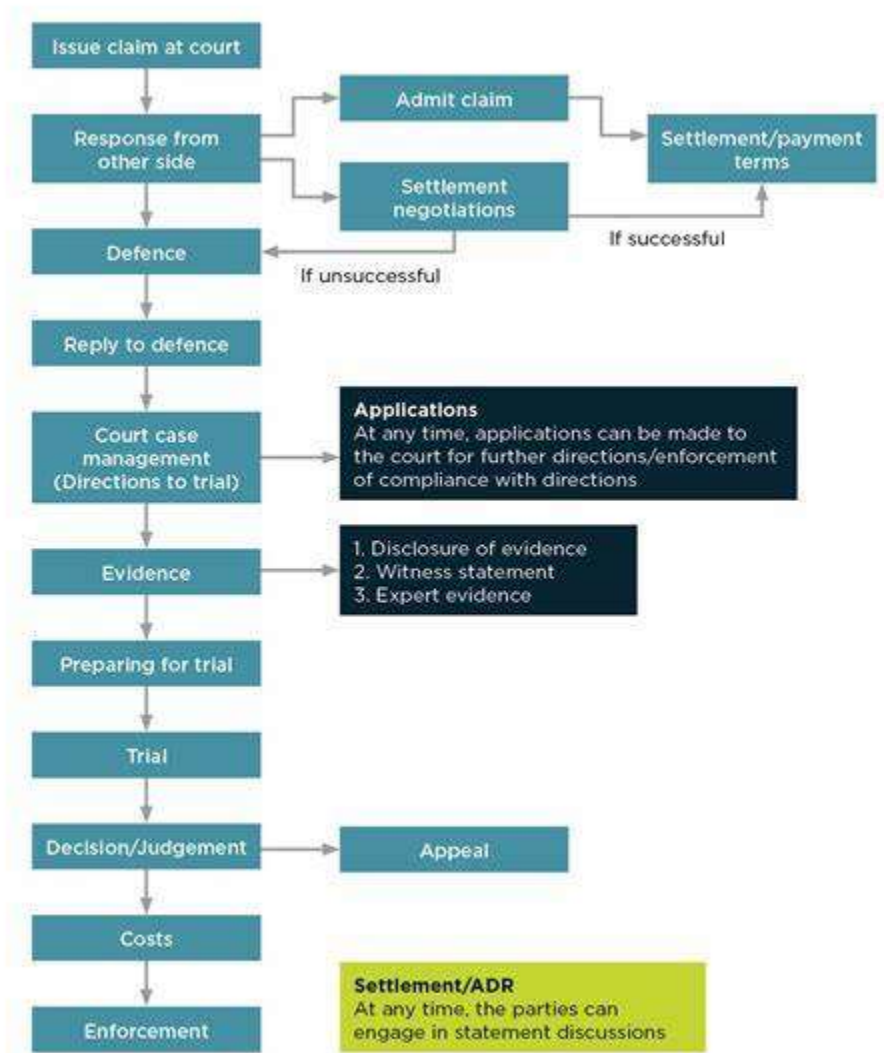


Figure 4.2.2. class Diagram

4.2.3 Judiciary system sequence Diagram

A sequence diagram is an interaction diagram that details how operations are carried out, capturing interaction between objects in the context of collaboration (Visual Paradigm, 2019). They save time as they visualize the order of interaction.

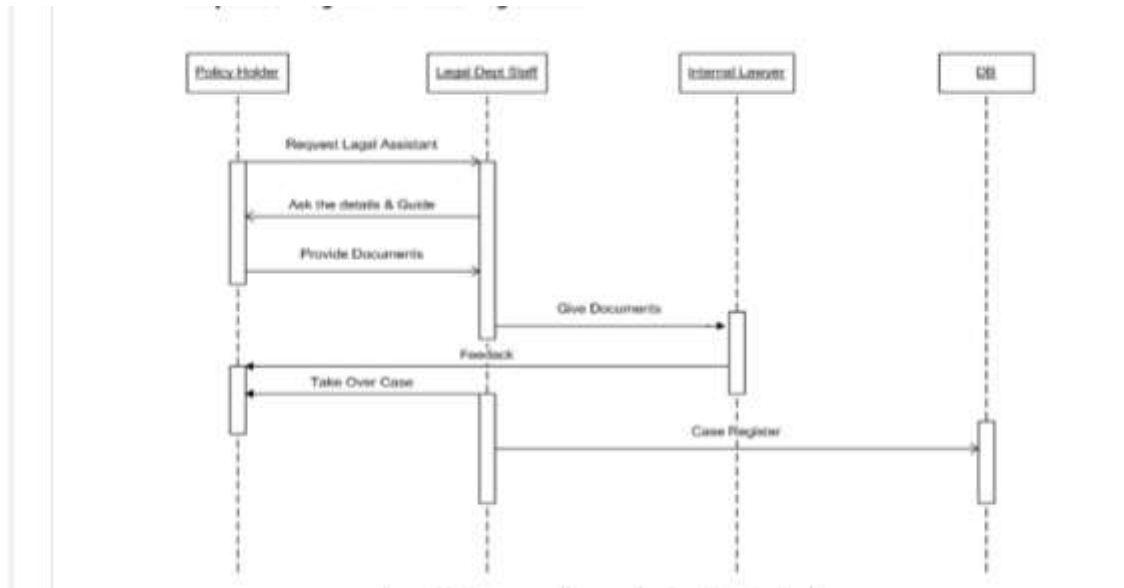


Figure 4.2.3 Judiciary system sequence diagram

4.2.4 Judiciary System Data Flow Diagram

A data flow diagram is a diagram that maps out the flow of information for any process or system (Lucidchart, 2017). The figure below shows a data flow diagram for the current judicial case filing system.

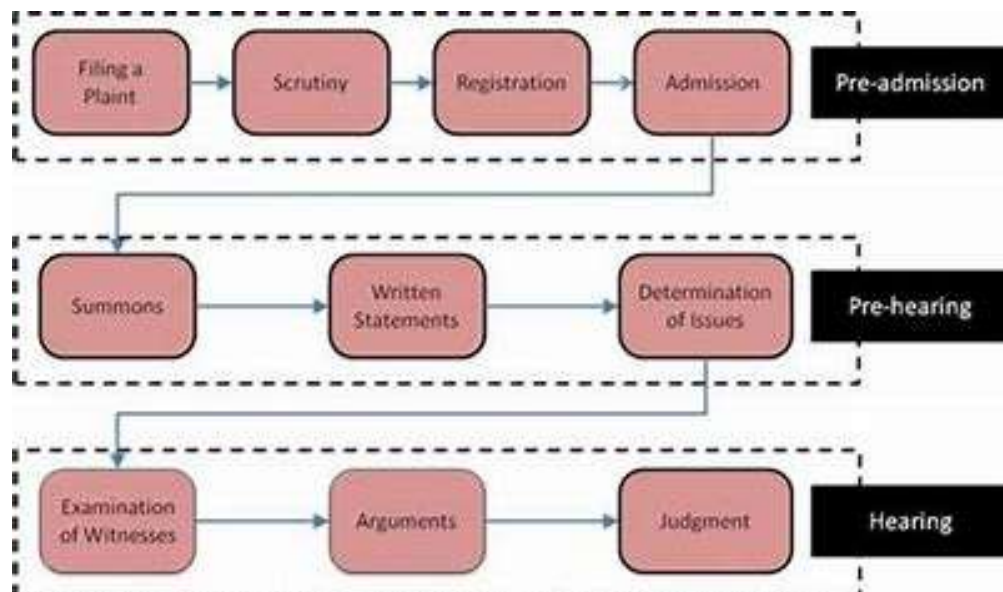


Figure 4.2.4 judiciary system Data flow diagram

4.2.5 Judiciary System Activity Diagram

A system activity diagram is

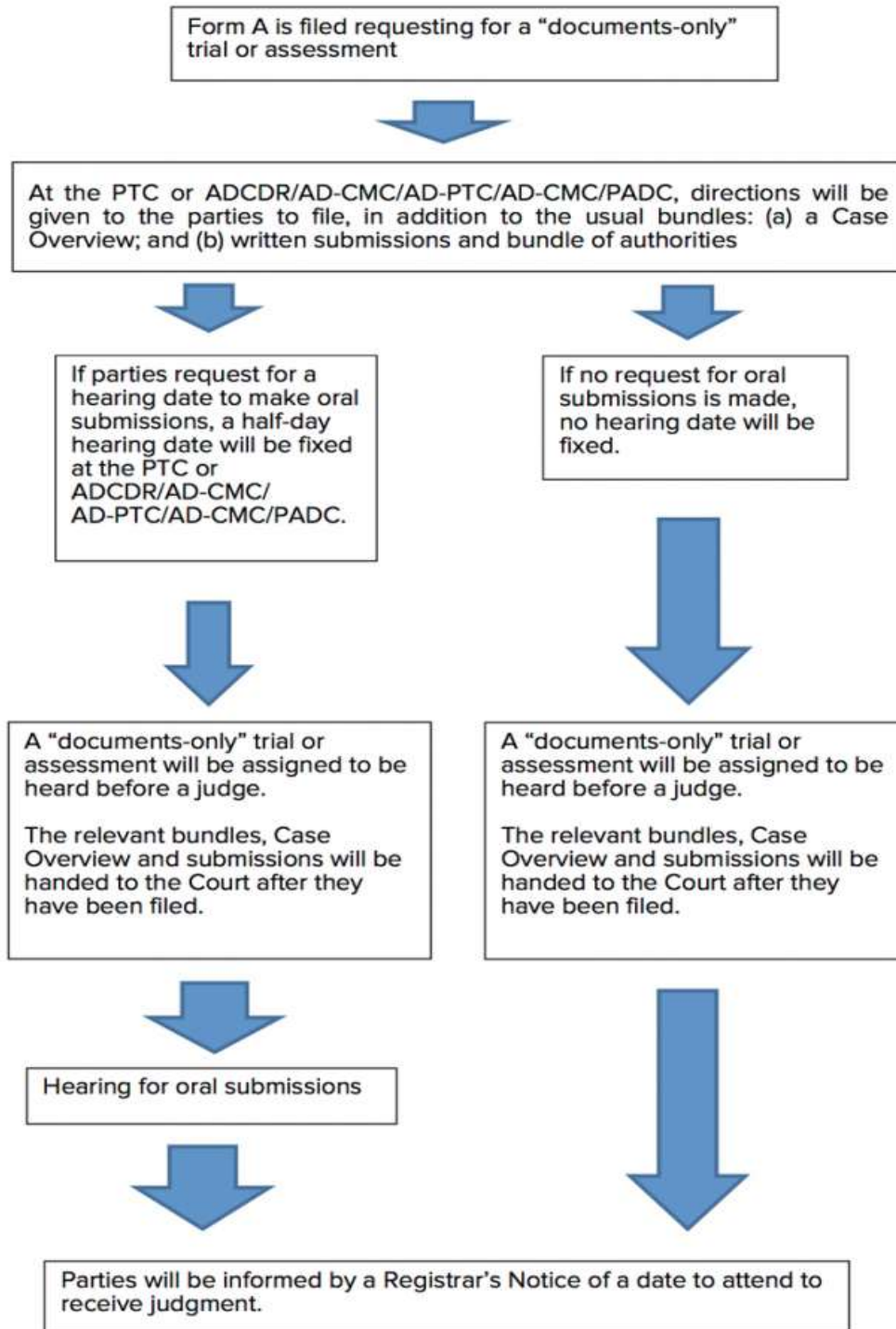


Figure 4.2.5 judiciary system activity diagram

4.3 Requirement gathering

Requirement gathering is the process of identifying your project's exact requirements from the start to the end (Asana, 2021).

4.3.1 Functional Requirements

Functional requirements are requirements which explain how the system work (Gorbachenko, 2021). They are those that one can be able to show how and demonstrate.

Importance

Functional requirements describe system behavior under specific conditions.

4.3.1.1 LOG IN

In this file encryption tool one of the features to ensure that only legitimate users can encrypt and decrypt files, is the authentication feature which takes place in the log in form which requires that users must provide a valid username and password that already exists in the database so as to log in. The user interface for log in captures details shown in the figure 4.3.1 below

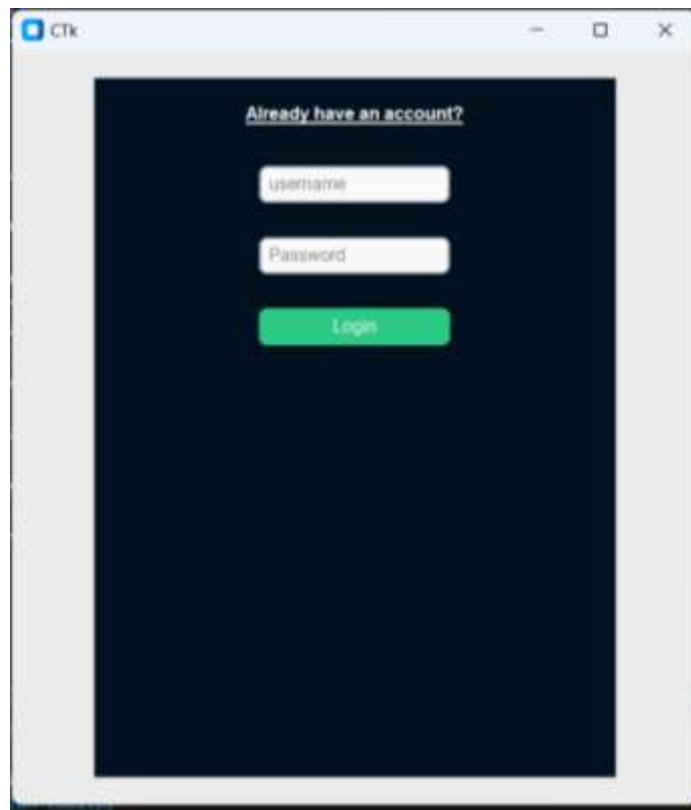


Figure 4.3.1.1 Log in page

```

69
70
71 def login():
72     frame1.destroy()
73     global frame2
74     frame2 = customtkinter.CTkFrame(app, bg_color='0001220',fg_color='0001220',width=470,height=360)
75     frame2.pack(pady=20, padx=60, fill="both", expand=True)
76
77     login_label=customtkinter.CTkLabel(frame2,font=font3,text='Already have an account?', text_color='ffff', bg_color='0001220')
78     login_label.pack(pady=12, padx=10)
79
80     global entry1
81     global entry2
82
83     entry1 = customtkinter.CTkEntry (master=frame2, placeholder_text="username")
84     entry1.pack(pady=12, padx=10)
85
86     entry2 = customtkinter.CTkEntry (master=frame2, placeholder_text="Password", show="*")
87     entry2.pack(pady=12, padx=10)

```

Figure 4.3.1.2 codes for log in

Figure 4.3.1.2 shows the programmers python codes of the log in interface.

Immediately the user keys in his or her username and password, the system will check in the database to see if the user exists and if not, it will display an error message that the user does not exist. However, if the username is for a registered user, and the user enters a wrong password, it will bring an error message that the user does not exist. However, if the details entered match the details in the database, the log in process will be successful, and the encryption tool will be automatically opened.

4.3.1.2 OPERATIONAL INTERFACE

In the encryption tool, the user now chooses which function he/she wants to perform with the tool from the list of buttons in the tool. Below is the interface of the tool in figure 4.3.2.1 below.

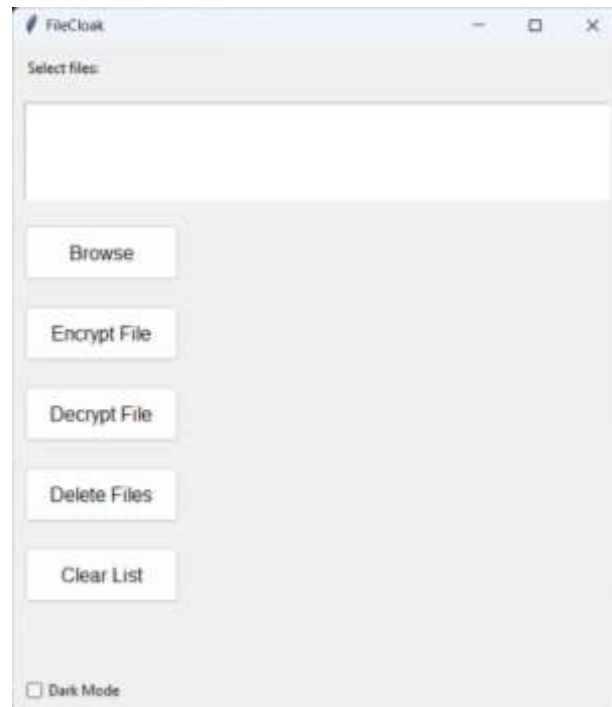


Figure 4.3.1.3 Encryption and decryption tool interface.

In order to encrypt a file the user will click on browse, then it will enable the user to choose a file of his/her choice from the computer being used. After selecting the target file, the user should click on the button “Encrypt file” in order to perform the encryption process.

In order to decrypt a file the user will click on browse, then it will enable the user to choose a file of his/her choice from the computer being used. After selecting the target file, the user should click on the button “Decrypt file” in order to perform the decryption process. Figure 4.3.2.2 below shows the programmers code of encryption and decryption interface.

```

# encryption and decryption tool.py
main.py

# Create the main window
app = tk.Tk()
app.title("FileCloak")
app.geometry("500x550") # set initial window size

# Create style for buttons
style = ttk.Style()

# Dark mode style
style.configure("Dark.Button", padding=10, font=("Arial", 12), foreground="white", background="#333333")

# light mode style
style.configure("Button", padding=10, font=("Arial", 12))

# Dark mode variable
dark_mode_var = tk.BooleanVar(value=False)

# Create widgets
File_label = tk.Label(app, text="Select Files:")
File_list = tk.Listbox(app, height=5, width=30)
browse_button = ttk.Button(app, text="Browse", command=browse_files)
encrypt_button = ttk.Button(app, text="Encrypt File", command=encrypt_file)
decrypt_button = ttk.Button(app, text="Decrypt File", command=decrypt_file)
delete_button = ttk.Button(app, text="Delete Files", command=delete_files)
clear_button = ttk.Button(app, text="Clear list", command=clear_file_list)
result_text = tk.Label(app, text="", fg="green", anchor="w")
dark_mode_checkbox = ttk.Checkbutton(app, text="Dark Mode", variable=dark_mode_var, command=toggle_dark_mode)

# Place widgets using grid layout
File_label.grid(row=0, column=0, sticky="w", padx=10, pady=10)
File_list.grid(row=1, column=0, padx=10, pady=10, columnspan=4, sticky="news")
browse_button.grid(row=3, column=0, padx=10, pady=10, sticky="w")
encrypt_button.grid(row=4, column=0, padx=10, pady=10, sticky="w")
decrypt_button.grid(row=5, column=0, padx=10, pady=10, sticky="w")
delete_button.grid(row=6, column=0, padx=10, pady=10, sticky="w")
clear_button.grid(row=7, column=0, padx=10, pady=10, sticky="w")
result_text.grid(row=8, column=0, columnspan=2, padx=10, pady=10, sticky="w")
dark_mode_checkbox.grid(row=9, column=0, columnspan=2, padx=10, pady=10, sticky="w")

```

Figure 4.3.1.4 Tools interface

4.3.2 Nonfunctional requirements

These are things that the user can't have directly but are operating in the background (Gorbachenko, 2021). In this system, the non-functional requirements are encryption, decryption and authentication. The figures 4.3.2.1, 4.3.2.2, and 4.3.2.3 show the three nonfunctional requirements respectively.

```

# encryption and decryption tool.py
main.py

def encrypt_file():
    file_to_encrypt = File_list.get(1, 0, "end").strip()
    try:
        with open(file_to_encrypt, "rb") as f:
            file_data = f.read()
            encrypted_data = fernet.encrypt(file_data)

            encrypted_file = file_to_encrypt + ".enc"
            with open(encrypted_file, "wb") as f:
                f.write(encrypted_data)

            # Delete the original file
            os.remove(file_to_encrypt)

            result_text.config(text=f"encrypted and deleted {file_to_encrypt}", fg="green")
    except Exception as e:
        result_text.config(text=f"An error occurred: {e}", fg="red")

```

Figure 4.3.2.1 encryption

```

103
104 def decrypt_file():
105     file_to_decrypt = file_list.get(1.0, "end-1c").strip()
106     try:
107         with open(file_to_decrypt, "rb") as f:
108             file_data = f.read()
109             decrypted_data = fernet.decrypt(file_data)
110
111             decrypted_file = file_to_decrypt.replace(".enc", "")
112             with open(decrypted_file, "wb") as f:
113                 f.write(decrypted_data)
114
115             result_text.config(text=f"Decrypted {file_to_decrypt} to {decrypted_file}", fg="green")
116     except Exception as e:
117         result_text.config(text=f"An error occurred: {e}", fg="red")
118

```

Figure 4.3.2.2 decryption

```

Run Terminal Help. FileClock+loginsystem
le-encryption and decryption tool.py X main.py
le-encryption and decryption tool.py > _
|
def login_account():
    username = entry1.get()
    password = entry2.get()
    if username != "" and password != "":
        cur.execute("SELECT password FROM nowdata WHERE username=?", [username])
        result = cur.fetchone()
        if result:
            if bcrypt.checkpw(password.encode("utf-8"), result[0]):
                frame2.destroy()
                messagebox.showinfo("Success", "logged in successfully.")
                app.destroy()
                call(["python", "main.py"])
            else:
                messagebox.showerror("Error", "Invalid Password")
        else:
            messagebox.showerror("Error", "Username does not Exist.")
    else:
        messagebox.showerror("Error", "Enter all data")

```

Figure 4.3.2.3 authentication using password

4.4 Summary

The file encryption and decryption tool has been tested and proven to be feasible in the various areas in terms of legal, operational and economic perspective. The system has been described in detail using case diagram, sequence diagram, Data Flow Diagram and activity diagram that shows all the processes involved. These diagrams elaborate the system functionality and operations. The system also has functional and non-functional requirements like logins, encryption and decryption which make it to be more efficient in its operations.

Appendix

Questionnaire

Directions: please answer truthfully and follow the guidelines given. Tick where the answer is either yes or no and write presentable answers in the explanatory questions. Confidentiality is exercised when handling the questionnaires

1. How secure is your system in terms of privacy and confidentiality of sensitive data

2. Would you like to install a file encryption and decryption tool in your system
Yes ☐ No ☐
3. Is there an existing encryption and decryption tool in your team?
Yes ☐ No ☐
4. Is there a system that ensures that data privacy and confidentiality throughout and if yes have you being able to interact with it?

5. What percentage of data do you feel is encrypted in your computer?

6. What percentage of data is present in the recycle bin in your computer?

7. According to the percentage of data you found in your recycle bin, what amount of data do you feel has been securely deleted since you started using the organizations computer?

8. What amount of sensitive data in the computer allocated to you is in plaintext (readable form and unencrypted)?

9. Is there a management team and plan in place to deal with the confidentiality issue?
Yes ☐ No ☐
10. What kind of system will you recommend to be developed? _____

Feasibility study report for file encryption tool in the Judiciary system set up

The purpose of this feasibility report is to address Purpose

the challenges on how courts in the judiciary suffer data leakage or data loss from both insider and external attacks. This has led to loss of reputation to the Judiciary and loss of confidence in the modern case solving system. This report investigated three possible solutions to make sure that the problem is solved.

First solution

The first solution is to develop a tool that enables court clerks, court administrators and/or any staff handling a softcopy of the casefile either in the registry or courtroom to store them in an encrypted form, for as long as the casefile is in the computer folders.

Economic feasibility

The file encryption and decryption tool is an economy friendly tool as it is created using lines codes. Therefore, the only economic cost that the company will incur will be labor cost for the programmer which will be ksh 50,000 monthly for 3 months. A total of ksh 150,000. The system shall need one annual maintenance costing ksh 60,000 and the service can go up to 5years.

Operational feasibility

This solution will greatly help judiciary operations to run smoothly because there will not be cases of missing files during times of hearings which cause the innocent party to feel as if there is some corruption being done to make the file disappear.

Technical feasibility

This solution does not require any additional hardware or software but the IT personnel of the Judiciary station should be well trained on how the system will work and make use of the available software

Legal Feasibility

This solution meets all legal and contractual laws of a country. There are no any violations of the law that comes out of the file encryption and decryption tool using fernet algorithm.

Second solution

The second solution is not to make any changes to the judiciary system but just install the file encryption and decryption tool.

Economic feasibility

This solution won't incur the judiciary any cost. This is because there are no more additional hardware and software tools to make it expensive.

Operational feasibility

This solution is not expected to change on how the judiciary system procedures were done.

Technical feasibility

System implementation will not require any new hardware and software for it to work.

Legal feasibility

This system meets all the country's contractual and legal laws.

Recommendation

After conducting the feasibility study, the first solution is recommended mainly because it will impact the Judiciary system. The judiciary and its stations will not suffer financially and it will not also lose important information belonging to people seeking justice and also those of criminals that can be used as evidence.

Research Schedule

The research schedule was presented in a form of Gantt chart. The gantt chart shows all the tasks that were carried out in the course of carrying the project. An estimate of the duration of each task in weeks was done

ACTIVITIES	SEPT	OCT	NOV	DEC	JAN	FEB	MAR
CHAPTER ONE INTRODUCTION							
CHAPTER TWO LITERATURE REVIEW							
CHAPTER THREE METHODOLOGY							
CHAPTER FOUR							
CHAPTER FIVE							
CHAPTER SIX							

CHAPTER SEVEN							
---------------	--	--	--	--	--	--	--

Budget

COMPONENTS	QUANTITIES	COST (Kshs)
Laptop	1	25,000
PRINTING	40 PAGES	200
BINDING	1 DOCUMENT	50
INTERNET BUNDLES	20GB	1000
	TOTALS	26,250

REFERENCES

Bhandari, P. (2021, October 18). *A Guide to Ethical Considerations in Research*. Scribbr.

<https://www.scribbr.com/methodology/research-ethics/>

Bless, G. (2023, March 11). *Challenges and limitations of encryption*. Medium. [https://gupta-](https://gupta-bless.medium.com/challenges-and-limitations-of-encryption-33b79ed046d4#:~:text=Encrypted%20data%20may%20not%20be)

[bless.medium.com/challenges-and-limitations-of-encryption-](https://gupta-bless.medium.com/challenges-and-limitations-of-encryption-33b79ed046d4#:~:text=Encrypted%20data%20may%20not%20be)

[33b79ed046d4#:~:text=Encrypted%20data%20may%20not%20be](https://gupta-bless.medium.com/challenges-and-limitations-of-encryption-33b79ed046d4#:~:text=Encrypted%20data%20may%20not%20be)

DeBose, K. (2018, September 21). *Research Guides: Research Methods Guide: Research Design*

& Method. Guides.lib.vt.edu. [https://guides.lib.vt.edu/researchmethods/design-](https://guides.lib.vt.edu/researchmethods/design-method#:~:text=Research%20design%20is%20a%20plan)

[method#:~:text=Research%20design%20is%20a%20plan](https://guides.lib.vt.edu/researchmethods/design-method#:~:text=Research%20design%20is%20a%20plan)

Duggal, N. (2020, October 21). *What Is Data Processing: Types, Methods, Steps of Data*

Processing Cycle. Simplilearn.com. [https://www.simplilearn.com/what-is-data-](https://www.simplilearn.com/what-is-data-processing-article)

[processing-article](https://www.simplilearn.com/what-is-data-processing-article)

Ewen, A. (2018). *Network Workshop Series D E V E L O P I N G E F F E C T I V E D A T A C O L L E C T I O N T O O L S*. [https://www.bumc.bu.edu/crrro/files/2018/12/Dec-RPN-Developing-](https://www.bumc.bu.edu/crrro/files/2018/12/Dec-RPN-Developing-Effective-Data-Collection-Tools.pdf)

[Effective-Data-Collection-Tools.pdf](https://www.bumc.bu.edu/crrro/files/2018/12/Dec-RPN-Developing-Effective-Data-Collection-Tools.pdf)

Hitachi. (2023). *System development methodology : Research & Development : Hitachi*.

Www.hitachi.com.

https://www.hitachi.com/rd/glossary/s/system_development_methodology.html#:~:text=Explanation

IBM. (2021, March 4). Use-case diagrams. Wwww.ibm.com.

<https://www.ibm.com/docs/en/rational-soft-arch/9.6.1?topic=diagrams-use-case>

Indeed Editorial Team. (2023, August 21). *What Is a Software Development Methodology? (With 11 Types)*. Indeed Career Guide. [https://au.indeed.com/career-advice/career-](https://au.indeed.com/career-advice/career-development/software-development-methodology)

[development/software-development-methodology](https://au.indeed.com/career-advice/career-development/software-development-methodology)

Indeed Editorial Team. (2022, June 3). Use Case Diagram: Definition and Examples. Indeed Career Guide. [https://www.indeed.com/career-advice/career-development/use-case-](https://www.indeed.com/career-advice/career-development/use-case-diagram)

[diagram](https://www.indeed.com/career-advice/career-development/use-case-diagram)

Ivanovs, A. (2023). 8 best tools for cryptography and encryption. *Stack Diary*.

<https://stackdiary.com/tools-for-cryptography-and-encryption/>

Kenton, W. (2023, March 17). Considering a New Venture? Then Consider a Feasibility Study.

Investopedia. <https://www.investopedia.com/terms/f/feasibility-study.asp>

Kumarurpadhyay, R. (2020, May 5). *Advantages and Disadvantages of using Spiral Model*.

GeeksforGeeks. <https://www.geeksforgeeks.org/advantages-and-disadvantages-of-using-spiral-model/>

Legal Feasibility of Projects. (n.d.). Daad & Kherad Law Firm. Retrieved January 29, 2024, from

<https://www.dandk.ir/en/special-services/legal-feasibility-of-projects/>

Lucidchart. (2017). What is a Data Flow Diagram | Lucidchart. Lucidchart.com.

<https://www.lucidchart.com/pages/data-flow-diagram>

Luxton, D. D., Kayl, R. A., & Mishkind, M. C. (2012). mHealth data security: The need for HIPAA-compliant standardization. *Telemedicine and e-Health*, 18(4), 284-288.

Momoh, O. (2023, April 29). *Understanding Populations in Statistics and Everyday Use*.

Investopedia.

<https://www.investopedia.com/terms/p/population.asp#:~:text=A%20population%20is%20the%20complete>

Orishaba, G. (2022). *An encryption and decryption tool for privacy and confidentiality of patients' data in electronic health records in Uganda* (Doctoral dissertation, Makerere university).

Project, T. P. of the G. (2020, January 7). *The GNU Privacy Guard*. Www.gnupg.org.

<https://www.gnupg.org/>

Roach, J. (2023, October 23). *VeraCrypt Review - Updated 2020*. Cloudwards.

<https://www.cloudwards.net/veracrypt-review/>

Recovery of Encryption Keys from Memory Using a Linear Scan. (2008, March 1). IEEE Conference Publication | IEEE Xplore.

<https://ieeexplore.ieee.org/abstract/document/4529504>

Stedman, C. (2022). *What is data collection? - Definition from WhatIs.com*. SearchCIO.

<https://www.techtarget.com/searchcio/definition/data-collection>

Synopsis. (2023). *What Is the Software Development Life Cycle (SDLC) and How Does It*

Work? | Synopsys. Www.synopsys.com. <https://www.synopsys.com/glossary/what-is->

[sdlc.html#:~:text=The%20Software%20Development%20Life%20Cycle%20\(SDLC\)%20is%20a%20structured%20process](#)

TENDER, N. PROVISION OF CONSULTANCY SERVICES FOR CERTIFICATION ON
ISO/IEC 27001: 2022

THE FULLSTORY TEAM. (2023, October 17). *Qualitative vs. Quantitative Data: What's the Difference? And Why They're So Valuable* | FullStory. Wwww.fullstory.com.

<https://www.fullstory.com/blog/qualitative-vs-quantitative-data/#:~:text=Quantitative%20research%20methods%20are%20measuring>

Tresorit Team. (2023, August 16). *History of Encryption: Roots of Modern-Day Cybersecurity*.

Tresorit Blog. Retrieved October 3, 2023, from <https://tresorit.com/blog/the-history-of-encryption-the-roots-of-modern-day-cyber-security/#:~:text=The%20first%20recorded%20instance%20of,to%20be%20sent%20and%20received>

Tuovila, A. (2020, May 28). *Sampling*. Investopedia.

<https://www.investopedia.com/terms/s/sampling.asp>.

Visual Paradigm. (2019). What is Sequence Diagram? Visual-Paradigm.com.

<https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-sequence-diagram/>

Zhang, Q., Jia, S., Chang, B., & Chen, B. (2018). Ensuring data confidentiality via plausibly deniable encryption and secure deletion—a survey. *Cybersecurity*, 1(1), 1-20.

